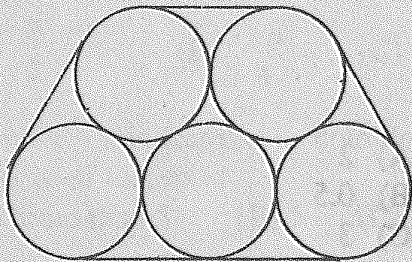


Junior level Screening Test - 1991.

1. Five circular logs of wood each of diameter 1m are held together by a taut rope as shown. The length in metres of the rope is

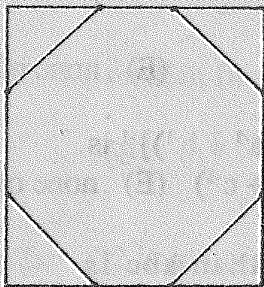


- (A) $5 + \pi$
 (B) 5
 (C) $5 + \frac{\pi}{2}$
 (D) $\frac{5 + \pi}{2}$
 (E) $10 + \pi$

2. The wheels of a truck travelling at 60 km/hr. make 4 revolutions per second. The diameter of each wheel in metre is

- (A) $\frac{25}{12\pi}$ (B) $\frac{6\pi}{25}$ (C) $\frac{25\pi}{6}$ (D) $\frac{100}{6\pi}$ (E) $\frac{25}{6\pi}$

3. A regular octagon is formed by cutting off corners of a square as shown in the diagram. If the side of the square is n cms., the total area removed in sq. cms. is



- (A) $4(n - \sqrt{2})$
 (B) $4\sqrt{2}n^2$
 (C) $(n^2 - 4n)/2$
 (D) $n^2(3 - 2\sqrt{2})$
 (E) none of the above

4. The coefficient of x^{99} in the expansion

$(x - 1)(x - 2) \dots (x - 100)$ is

- (A) -5050 (B) -4950 (C) -100 (D) -99 (E) -4851

5. If all the roots of $x^4 - 8x^3 + 24x^2 + bx + c = 0$ are real and positive, then the value of b is

- (A) -24 (B) -32 (C) 0 (D) -4 (E) none of the above

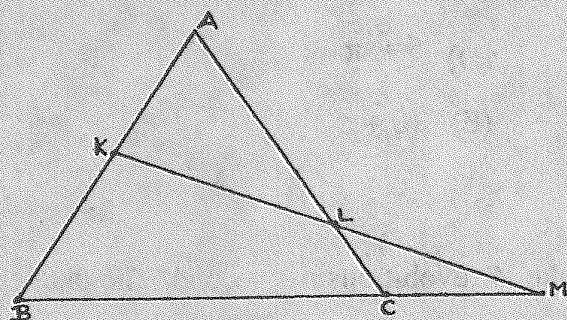
6. The difference between the larger root and the smaller root of $x^2 - px + \frac{p^2 - 1}{4} = 0$ is

- (A) 0 (B) 1 (C) 2 (D) p (E) $(p + 1)$

7. The value of $\sqrt{2\sqrt{7} - 8} - \sqrt{8 + 2\sqrt{7}}$ is

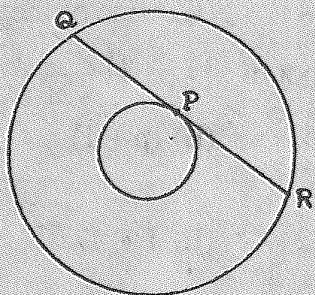
- (A) 2 (B) 1 (C) -1 (D) -2 (E) $1 + \sqrt{7}$

8. 852 digits are used to number the pages of a book consecutively from page 1. The number of pages in the book is
 (A) 320 (B) 247 (C) 316 (D) 251 (E) 284
9. The five tyres of a car (four road tyres and one spare) were each used equally in a journey of 20,000 kms. The number of kms. of use of each tyre was
 (A) 4000 (B) 5000 (C) 16000 (D) 20000 (E) 100000
10. In the figure $AB = AC$ and $KL = 2LM$. KB/LC is



- (A) 2
 (B) 0.5
 (C) 3
 (D) $\frac{1}{3}$
 (E) none of the above

11. ABC is a triangle right angled at A. If $AB = 4$, $AC = 3$, then the altitude AD from A is
 (A) 12 (B) $\frac{3}{4}$ (C) $\frac{4}{3}$ (D) 2.4 (E) none of the above
12. If $xy : yz : zx = 2 : 3 : 4$, then $\frac{y}{zx} : \frac{z}{xy}$ is
 (A) 2 : 1 (B) 1 : 4 (C) 4 : 1 (D) 1 : 2 (E) none of the above
13. The value of $\{1 + a(b^{-1} + c^{-1})\}^{-1} + \{1 + b(c^{-1} + a^{-1})\}^{-1} + \{1 + c(a^{-1} + b^{-1})\}^{-1}$ is
 (A) $3 + a^{-1} + b^{-1} + c^{-1}$ (B) 3 (C) 1 (D) $3(1 + a^{-1} + b^{-1} + c^{-1})$ (E) none of the above
14. If $a = \frac{1}{3} \log_3 \left(\frac{1}{5}\right)$, $b = \frac{1}{3} \log_5 \left(\frac{1}{7}\right)$, $c = \frac{1}{7} \log_7 \left(\frac{1}{3}\right)$, then abc is
 (A) 1 (B) -1 (C) -105 (D) $-\frac{1}{105}$ (E) none of the above
15. The tangent at P to the smaller circle of two concentric circles intersect the outer circle at Q, R. The length of QR is 14cm. The area of the annulus in sq. cm. is



- (A) 196π
 (B) 49π
 (C) $49\pi/4$
 (D) 49
 (E) not determined by the given information

16. A regular dodecagon (12 sides) is inscribed in a circle with radius r cms. The area of the dodecagon in square cms. is
 (A) $3r^2$ (B) $2r^2$ (C) $\frac{3r^2\sqrt{3}}{4}$ (D) $\sqrt{3}r^2$ (E) $3\sqrt{3}r^2$
17. If $a = \log_8 225$, $b = \log_2 15$ then a in terms of b is
 (A) $\frac{b}{2}$ (B) $\frac{2b}{3}$ (C) b (D) $\frac{3b}{2}$ (E) $2b$
18. When $x^9 - x$ is factored as completely as possible, into polynomials with integral coefficients the number of factors is
 (A) more than 5 (B) 5 (C) 4 (D) 3 (E) 2
19. The bisectors of the exterior angles at B and C of a triangle ABC meet at D. The angle $\angle BDC$ in degrees is
 (A) $\frac{1}{2}(90 - A)$ (B) $90 - A$ (C) $\frac{1}{2}(180 - A)$ (D) $180 - A$ (E) $180 - 2A$
20. If the radius of a circle is increased by 1 unit, the ratio of the new circumference to the new diameter is
 (A) $\pi + 2$ (B) $\frac{2\pi + 1}{2}$ (C) π (D) $\frac{2\pi - 1}{2}$ (E) $\pi - 2$
21. In $\triangle ABC$, the median from A is perpendicular to the median from B. If $BC = 7$, $AC = 6$ then AB is
 (A) 4 (B) $\sqrt{17}$ (C) 4.25 (D) $2\sqrt{5}$ (E) 4.5
22. The number 432 is to be written in factorial base of numeration i.e. $432 = a_1 + a_2 \cdot 2! + \dots + a_n \cdot n!$ where $a_1, a_2 \dots a_n$ are integers such that $0 \leq a_k \leq k$ and $k! = 1 \cdot 2 \dots (k - 1) \cdot k$, $k = 1, 2, 3 \dots$. Then a_4 is
 (A) 0 (B) 1 (C) 2 (D) 3 (E) none of the above
23. Each side of a $\triangle ABC$ is 12 cm. D is the foot of the perpendicular from A on BC and E is the mid point of AD. The length of BE in cm is
 (A) $\sqrt{18}$ (B) $\sqrt{28}$ (C) 6 (D) $\sqrt{63}$ (E) $\sqrt{98}$
24. If $\log_{10} 2 = a$ and $\log_{10} 3 = b$ then $\log_5 12$ is
 (A) $\frac{a+b}{1+a}$ (b) $\frac{2a+b}{1+a}$ (c) $\frac{a+2b}{1+a}$ (d) $\frac{2a+b}{1-a}$ (e) $\frac{a+2b}{1-a}$

25. Let r, s be the roots of $ax^2 + bx + c = 0$. The equation with roots $ar + b$ and $as + b$ is
- (A) $x^2 - bx - ac = 0$ (B) $x^2 - bx + ac = 0$
 (C) $x^2 + 3bx + ca + 2b^2 = 0$ (D) $x^2 + 3bx - ca + 2b^2 = 0$
 (E) none of the above
26. The last digit of 2137^{753} is
- (A) 1 (B) 3 (C) 5 (D) 7 (E) 9
27. Let S be the set consisting of the squares of all positive integers. If a certain operation on one or more members of the set always yields a member of the set, we say that the set is closed under that operation. S is closed under
- (A) addition (B) multiplication (C) division (D) finding positive integral root
 (E) none of the above
28. The diagonal of a cube is $\sqrt{12}$ cm. The volume of the cube in cubic cms is
- (A) 8 (B) 12 (C) 24 (D) $3\sqrt{12}$ (E) $12\sqrt{12}$
29. A certain substance doubles its volume every minute. At 9 : 00 am a small amount is placed in a container and at 10 : 00 am the container just fills. The time at which the container was quarter full was
- (A) 9 : 15 am (B) 9 : 30 am (C) 9 : 45 am (D) 9 : 50 am (E) 9 : 58 am
30. For a real number x , $[x]$ denotes the integral part of x . That is $[x]$ is the largest integer less than or equal to x . For example, $[1.24] = 1$, $[-1.24] = -2$. The value of
- $$\left[\frac{1}{2}\right] + \left[\frac{1}{2} + \frac{1}{100}\right] + \left[\frac{1}{2} + \frac{2}{100}\right] + \dots + \left[\frac{1}{2} + \frac{99}{100}\right]$$
- is
- (A) 49 (B) 50 (C) 48 (D) 51 (E) none of the above

